Appendix. Search strategies.

Cochrane Library CENTRAL search strategy

Date Run: 20/12/2021 02:52:03

Comment:

ID Search Hits

#1 MeSH descriptor: [Stem Cells] explode all trees877

#2 MeSH descriptor: [Mesenchymal Stem Cells] explode all trees 159

#3 MeSH descriptor: [Cell- and Tissue-Based Therapy] explode all trees 6519

#4 ("stem cell*" or "cell therapy"):ti,ab,kw 12939

#5 #1 or #2 or #3 or #4 17489

#6 MeSH descriptor: [COVID-19] explode all trees 918

#7 MeSH descriptor: [SARS-CoV-2] explode all trees 599

#8 ("coronavirus infection" or "coronavirus disease" or "COVID-19" or "SARS-CoV-2"):ti,ab,kw 8322

#9 #6 or #7 or #8 8322

#10 #5 and #9 88

Database: OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to present

Search Strategy:

---

1 exp Stem Cells/ (239580)

2 exp Mesenchymal Stem Cells/ (44499)

3 exp Cell-/ and Tissue-Based Therapy.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (4666)

4 ("stem cell*" or "cell therapy").mp. (411572)

5 1 or 2 or 3 or 4 (428986)

6 exp COVID-19/ (127019)

7 exp SARS-CoV-2/ (100426)

8 ("coronavirus infection" or "coronavirus disease" or "COVID-19" or "SARS-CoV-2").mp. (208781)

9 6 or 7 or 8 (208781)

10 5 and 9 (1213)

****

Database: Embase <1974 to 2021 December 17>

Search Strategy:

---

1 exp Stem Cells/ (409994)

2 exp Mesenchymal Stem Cells/ (60464)

3 exp Cell-/ and Tissue-Based Therapy.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword heading word, floating subheading word, candidate term word] (113)

4 ("stem cell*" or "cell therapy").mp. (653754)

5 1 or 2 or 3 or 4 (675009)

6 exp COVID-19/ (169775)

7 exp SARS-CoV-2/ (50582)

8 ("coronavirus infection" or "coronavirus disease" or "COVID-19" or "SARS-CoV-2").mp. (219632)

9 6 or 7 or 8 (220576)

10 5 and 9 (2337)

****

Results of the (a priori) random-effects and (a posteriori) fixed-effects models for comparing the mortality of the MSC therapy to the standard treatment in patients with severe/critical COVID-19.

The priori:

DerSimonian-Laird random-effects method selected (MH Odds-Ratio)

Study	Effect	[95% Conf.	Interval]	% Weight
Kaushal 2020	0.459	0.088	2.395	8.73
Leng 2020	0.212	0.007	6.816	1.98
Meng 2020	.	.	.	6.20
Feng 2021	.	.	.	6.20
Häberle-2021	0.360	0.037	3.526	4.57
Xu 2021	0.231	0.042	1.275	8.15
Shu 2020	0.262	0.013	5.244	2.65
Adas 2021	0.500	0.097	2.577	8.85
Dilogo 2021	0.625	0.229	1.706	23.61
Lanzoni 2021	0.286	0.049	1.666	7.65
shi 2021	.	.	.	6.20
Ventura-Carmenate 20	0.686	0.175	2.683	12.79
Zhu 2021	0.186	0.008	4.240	2.44
Overall effect (dl)	0.518	0.318	0.844	100.00

Effects have been exponentiated

The posteriori:

Fixed-effect method selected (MH Odds-Ratio)

Study	Effect	[95% Conf.	Interval]	% Weight
Kaushal 2020	0.459	0.088	2.395	8.73
Leng 2020	0.212	0.007	6.816	1.98
Meng 2020	.	.	.	6.20
Feng 2021	.	.	.	6.20
Häberle-2021	0.360	0.037	3.526	4.57
Xu 2021	0.231	0.042	1.275	8.15
Shu 2020	0.262	0.013	5.244	2.65
Adas 2021	0.500	0.097	2.577	8.85
Dilogo 2021	0.625	0.229	1.706	23.61
Lanzoni 2021	0.286	0.049	1.666	7.65
shi 2021	.	.	.	6.20
Ventura-Carmenate 20	0.686	0.175	2.683	12.79
Zhu 2021	0.186	0.008	4.240	2.44
Overall effect (fe)	0.518	0.318	0.844	100.00

Effects have been exponentiated